

Quantum Machine Learning for Pneumonia Detection in Chest X-Ray Imaging: an Illustrative Case Study

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Abstract

Deep convolutional neural networks (CNNs) have become the de facto standard for automated pneumonia screening from chest X-ray (CXR) images, routinely reporting diagnostic accuracies above 85%. This performance, however, is obtained at the cost of classification heads with hundreds of thousands of trainable parameters, motivating interest in parameter-efficient alternatives. Variational quantum circuits (VQCs), operating within the constraints of Noisy Intermediate-Scale Quantum (NISQ) hardware, have been proposed as compact decision layers that can be appended to classical feature extractors. Beyond the pure VQC paradigm, quantum convolutional neural networks (QCNNs) and quantum kernel methods have been explored as complementary architectural routes. This paper reviews the computational and algorithmic foundations of such hybrid quantum–classical architectures for CXR-based pneumonia detection, covering classical CNN baselines, quantum data-encoding strategies (angle, amplitude, IQP feature maps, and data re-uploading), variational and convolutional ansatz design, quantum kernel construction, and the barren-plateau phenomenon that limits the trainability of deep parametrized circuits together with its principal mitigation strategies. As a concrete illustration, we describe and analyze a hybrid pipeline that couples a frozen ConvNeXt-Tiny feature extractor with nonlinear autoencoder dimensionality reduction (768→64) and a 6-qubit, 54-parameter data re-uploading VQC classifier, evaluated on a publicly available pediatric CXR dataset under an explicit train/test class-prevalence shift. The hybrid model attains a test AUC-ROC of 0.860 (versus 0.940 for a classical multilayer perceptron trained on identical features) while using approximately $39\times$ fewer trainable parameters in its decision layer (54 versus 2,113). We situate this case study within a broader survey of related hybrid approaches for respiratory and cardiac CXR analysis reported in the literature, and ground a discussion of the algorithmic trade-offs, evaluation methodology, and open problems—including state-preparation overhead, barren plateaus, dataset shift, and the absence, at present circuit scales, of a demonstrated computational quantum advantage. A dedicated section is reserved for forthcoming experiments on physical quantum hardware with error mitigation; its completion will allow direct quantification of the simulator-to-hardware performance gap.

Keywords: quantum machine learning; variational quantum circuits; hybrid quantum–classical neural networks; quantum convolutional neural networks; quantum kernel methods; pneumonia detection; chest X-ray classification; NISQ; barren plateaus; medical image analysis

1 Introduction

Pneumonia remains one of the leading causes of morbidity and mortality worldwide, with an estimated global burden in the range of millions of cases annually and disproportionate impact on children under five and immunocompromised adults [3, 12]. Chest radiography (CXR) is the most widely used first-line imaging modality for pneumonia screening because it is inexpensive, fast, and broadly available compared to computed tomography [12]. Manual interpretation of CXR images is subject to inter-observer variability, reader fatigue, and uneven access to trained radiologists, which has motivated three decades of research into computer-aided diagnosis and, more recently, deep-learning-based automated screening [21, 16].

CNNs, and in particular deep residual architectures such as ResNet, have demonstrated diagnostic performance on CXR pneumonia classification that is competitive with expert radiologists, typically exceeding 85% classification accuracy [10, 16, 32]. This performance, however, comes at a substantial computational cost: fully connected classification heads appended to CNN backbones commonly require on the order of 10^5 – 10^6 trainable parameters. The associated memory and energy costs are non-trivial for edge or point-of-care deployment scenarios and motivate the search for parameter-efficient decision layers.

In parallel, the maturation of NISQ devices has generated interest in variational quantum algorithms (VQAs) as parameter-efficient function approximators [29, 26]. A VQC parametrizes a family of unitary transformations $U(\theta)$ and, when combined with classical optimization of θ , can implement expressive decision boundaries using a number of trainable parameters that grows only linearly with circuit depth and qubit count [20, 2]. This has prompted a growing body of work on *hybrid quantum–classical* (HQC) architectures, in which a classical CNN performs high-dimensional feature extraction and a quantum component performs low-dimensional classification, confining the quantum computation to regimes tractable on current simulators or hardware.

A number of recent studies have applied HQC architectures specifically to thoracic imaging tasks. Kulkarni et al. [14] demonstrated that replacing a classical CNN layer with a VQC yields statistically significant improvements for pneumonia detection on CXR images. Rao et al. [28] reported a 98.1% test accuracy on a COVID-19 Radiography Dataset using a hybrid CNN–VQC pipeline deployed partly on an IBM Q simulator, while Decoodt et al. [6] achieved an AUC of up to 0.93 for cardiomegaly detection using a DenseNet-121 backbone augmented with a variational quantum layer. At the same time, attention has shifted beyond the “quantum classifier head” paradigm: quantum convolutional neural networks (QCNNs) designed to mimic spatial locality in quantum circuits [5, 36], and quantum kernel methods that compute high-dimensional inner products via quantum interference [9, 8], represent architecturally distinct alternatives that may better exploit the inductive biases of image data.

Despite this activity, empirical evidence quantifying the accuracy–efficiency trade-off of HQC pipelines on realistic medical imaging tasks—particularly under distribution shift between training and deployment data—remains limited. Most published studies report results on a single static train/test split, on noise-free simulators, and without explicit discussion of the state-preparation and trainability bottlenecks intrinsic to variational quantum computing [33].

1.1 Contributions and Scope

This paper has two purposes. First, it provides a structured, computationally oriented review of the building blocks required to construct hybrid quantum–classical pneumonia classifiers: classical CNN backbones, dimensionality reduction, quantum data-encoding strategies, variational and convolutional ansatz design, quantum kernel construction, and the barren-plateau problem including its principal mitigation strategies. Second, it presents a detailed case study—a ConvNeXt-Tiny/autoencoder/VQC pipeline evaluated on a public pediatric CXR dataset—that grounds the algorithmic discussion with explicit numbers and illustrates the current gap between classical and quantum-assisted approaches.

The clinical and radiological background is kept intentionally brief; the emphasis throughout is on encoding schemes, circuit design, trainability, computational complexity, and evaluation methodology, consistent with a computer-science- and quantum-technology-oriented readership. We deliberately avoid overstating the clinical readiness of the reviewed methods: at the qubit counts considered here, no computational quantum advantage is established, and the results should be interpreted as a feasibility and parameter-efficiency study rather than as a validated diagnostic tool. A separate section (Section 6) is reserved for forthcoming experimental results on physical quantum hardware; its inclusion reflects the acknowledged importance of closing the simulator-to-hardware gap and will complete the empirical contribution of this work.

The remainder of the paper is organized as follows. Section 2 introduces the clinical context and the classical and quantum computational preliminaries. Section 3 reviews classical CNN-based approaches to CXR pneumonia detection. Section 4 describes quantum data encoding, variational circuits, QCNs, quantum kernels, and the barren-plateau phenomenon. Section 5 surveys related HQC work for CXR analysis and presents the ConvNeXt-Tiny/autoencoder/VQC case study. Section 6 provides the placeholder for real-hardware experiments. Section 7 discusses experimental design and evaluation metrics. Section 8 enumerates open problems and limitations. Section 9 outlines future research directions, and Section 10 concludes.

2 Background

2.1 Clinical and Imaging Context

Pneumonia is an acute inflammatory condition of the pulmonary parenchyma, most commonly of bacterial or viral etiology, in which the alveoli fill with fluid or exudate, impairing gas exchange [12]. On a CXR, healthy air-filled lung tissue appears radiolucent (dark), whereas pneumonic consolidation, interstitial infiltrates, and air bronchograms appear as localized or diffuse radiopaque (light) regions [12]. Bacterial pneumonia (e.g., *Streptococcus pneumoniae*) typically presents with lobar consolidation, while viral pneumonia tends to produce more diffuse interstitial patterns [12]. This visual heterogeneity, combined with overlapping appearances with other conditions (pulmonary edema, atelectasis, fibrosis), contributes to inter-observer variability in manual reading and motivates automated decision support [21].

2.2 Classical Deep Learning for Medical Image Classification

Modern CXR classifiers are almost universally built on deep CNN backbones pretrained on large natural-image corpora (e.g., ImageNet) and subsequently fine-tuned or used as fixed feature extractors on medical imaging tasks. ResNet architectures address the vanishing-gradient and degradation problems that historically limited the trainability of very deep networks by introducing residual (skip) connections: instead of learning a direct mapping $H(x)$, each residual block learns a residual function $F(x)$ such that

$$H(x) = F(x) + x, \quad (1)$$

which allows gradients to propagate across many layers without attenuation [10]. ResNet-50 uses a bottleneck design in which each residual block consists of a 1×1 convolution (dimensionality reduction), a 3×3 convolution (spatial filtering), and a further 1×1 convolution (dimensionality restoration), substantially reducing parameter count relative to a naive stacking of 3×3 convolutions while preserving representational depth [10]. In a typical CXR pipeline, the network is used *headless*, i.e., without its final classification layer, producing a fixed-length embedding (commonly of dimension 2048 for ResNet-50) for each input image, which is then passed to a task-specific classifier.

2.3 Elements of Quantum Computation

A qubit is the basic unit of quantum information, represented as a normalized vector in a two-dimensional complex Hilbert space,

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle, \quad |\alpha|^2 + |\beta|^2 = 1, \quad (2)$$

where $\alpha, \beta \in \mathbb{C}$ are probability amplitudes and $\{|0\rangle, |1\rangle\}$ is the computational basis [22]. Geometrically, a pure single-qubit state can be represented as a point on the Bloch sphere,

$$|\psi\rangle = \cos\left(\frac{\theta}{2}\right) |0\rangle + e^{i\varphi} \sin\left(\frac{\theta}{2}\right) |1\rangle, \quad (3)$$

and training a variational circuit can be understood, at the single-qubit level, as searching for rotation angles that steer this state toward a measurement outcome associated with the desired class.

Quantum algorithms are composed of unitary *gates*. Single-qubit rotation gates R_x, R_y, R_z (or the three-angle rotation $\text{Rot}(\phi, \theta, \omega)$) rotate the state around the corresponding axis of the Bloch sphere and constitute the trainable degrees of freedom of a variational circuit; the Hadamard gate creates equal superpositions; and the two-qubit controlled-NOT (CNOT) gate is the standard primitive for generating *entanglement*, a distinctively quantum correlation between qubits that cannot be reproduced by any local, classical description of the joint system and that is conjectured to underlie the representational power of quantum models [22].

The *expressibility* of a parametrized circuit is a quantitative measure of how uniformly it can generate states over the Hilbert space, often defined via the deviation of the induced distribution from the Haar measure [35]. Expressibility and entanglement capacity jointly govern the circuit’s ability to represent complex functions and, as detailed in Section 4.5, its susceptibility to trainability issues.

2.4 The NISQ Era and Variational Quantum Algorithms

Contemporary quantum processors operate in what Preskill termed the Noisy Intermediate-Scale Quantum (NISQ) era: devices with tens to a few hundred physical qubits that are not yet fault-tolerant and remain subject to gate errors and decoherence [26]. Variational quantum algorithms are the dominant algorithmic paradigm for extracting practical value from NISQ hardware. A VQA defines a parametrized unitary $U(\theta)$, executes it on a (possibly classically simulated) quantum device, measures an observable, and updates θ using a classical optimizer—a hybrid quantum–classical optimization loop that keeps quantum resource requirements shallow while delegating gradient-based optimization to classical compute [29, 20]. Variational Quantum Classifiers (VQCs) are the supervised-learning instance of this paradigm and form the quantum component of the hybrid architectures discussed in this paper.

3 Classical Approaches to Pneumonia Detection from Chest X-Rays

The dominant classical paradigm for CXR-based pneumonia detection follows a transfer-learning recipe: (i) a CNN backbone pretrained on a large natural-image dataset is used, either fine-tuned end-to-end or frozen as a fixed feature extractor; (ii) a fully connected classification head, typically consisting of one or more dense layers followed by a sigmoid or softmax output, is trained on the target task; and (iii) standard image preprocessing (resizing, normalization with respect to the pretraining dataset’s channel statistics) is applied to bridge the domain gap between natural and medical images. ResNet, DenseNet, and, more recently, vision-transformer backbones have all been applied to the publicly available pediatric CXR pneumonia dataset originating from the Guangzhou Women and Children’s Medical Center, with reported accuracies generally exceeding 85% and F1 scores in the 0.85–0.92 range [16, 32, 21].

From a purely computational standpoint, the classification head is the component of the pipeline most directly comparable to a quantum decision layer. For a 64-dimensional input (e.g., after dimensionality reduction, see Section 5) and a single hidden layer of width 32, a standard multilayer perceptron (MLP) classification head requires

$$P_{\text{MLP}} = (64 \times 32 + 32) + (32 \times 1 + 1) = 2,080 + 33 = 2,113 \text{ parameters}, \quad (4)$$

illustrating that even a comparatively shallow classifier on a compressed feature space carries a non-trivial parameter budget. This observation is the principal motivation for exploring VQC-

based decision layers, whose parameter count scales as $O(nL)$ in the number of qubits n and circuit layers L , rather than with the width of a dense layer (Section 4).

Two further classical design choices are directly relevant to the hybrid architectures discussed below. First, class imbalance—pneumonia typically constitutes 70–75% of samples in public pediatric CXR datasets—motivates the use of weighted loss functions (e.g., weighted binary cross-entropy) rather than unweighted cross-entropy, to avoid a majority-class bias. Second, dimensionality reduction (via PCA, linear discriminant analysis, or univariate feature selection) is frequently required before any quantum processing step, since CNN embeddings (commonly 512–2048-dimensional) exceed the practical qubit budgets of NISQ-era hardware and simulators by orders of magnitude.

4 Quantum Machine Learning Approaches

4.1 Data Encoding Strategies

Before a classical feature vector $x \in \mathbb{R}^N$ can be processed by a quantum circuit, it must be mapped, via a *state-preparation* routine, onto a quantum state $|\psi(x)\rangle$ in an n -qubit Hilbert space. The choice of encoding is consequential: it determines the geometry of the induced kernel, the depth of the state-preparation circuit, and, as recent theoretical work has shown, the trainability of the subsequent variational circuit [13, 31].

Basis encoding maps a bit string directly onto computational basis states, requiring one qubit per bit of the (typically pre-discretized) classical input. It scales poorly for continuous, high-dimensional features and is rarely used for image-derived embeddings.

Angle (rotation) encoding assigns each classical feature x_i to a rotation angle on a dedicated qubit, e.g., $R_y(x_i)$, and therefore requires $n = N$ qubits for an N -dimensional feature vector. It has the practical advantage of a shallow, easily implementable preparation circuit and has been shown to produce more consistent results on real QPUs than more complex encodings in at least one comparative hardware study [27].

Amplitude encoding embeds a normalized vector $x \in \mathbb{R}^N$, $\|x\|_2 = 1$, into the amplitudes of an n -qubit state,

$$|\psi(x)\rangle = \sum_{i=0}^{N-1} x_i |i\rangle \in \mathcal{H}_{2^n}, \quad N = 2^n, \quad (5)$$

requiring only $n = \log_2 N$ qubits. This exponential compression is the primary motivation for its use in hybrid CNN–VQC pipelines, since it allows, e.g., a 64-dimensional PCA-reduced embedding to be encoded on just $\log_2 64 = 6$ qubits.

IQP and ZZ feature maps apply layers of classically controlled phase rotations—in particular the instantaneous quantum polynomial (IQP) circuit structure—to produce highly non-linear, problem-specific feature maps [9]. The resulting kernel $k(x, x') = |\langle \psi(x) | \psi(x') \rangle|^2$ can be exponentially hard to evaluate classically, lending potential computational interest to quantum kernel classifiers (Section 4.4). Empirically, IQP-based encodings have been observed to improve precision on some tasks while reducing recall and F_1 scores relative to angle encoding, and their performance is more sensitive to hardware noise [27].

Data re-uploading addresses the limited expressibility of a single encoding layer by repeatedly interleaving data-encoding gates with trainable rotation layers throughout the circuit, effectively allowing each qubit to “see” the input multiple times [24]. This strategy has been

shown to confer universal function-approximation properties to single qubits and provides a practical route to expressible, shallow circuits compatible with NISQ depth budgets.

The compression advantage of amplitude encoding is not free: in the general case, exact amplitude-encoding state preparation for an arbitrary N -dimensional vector requires a circuit of depth $O(N)$, i.e., *exponential* in the number of qubits, unless the vector possesses exploitable structure or approximate preparation is acceptable [1]. This “input bottleneck” is a widely recognized caveat in the QML literature, and claims of exponential quantum speedup based on amplitude encoding alone should be treated with caution [1, 2]. In practice, most simulation-based studies—including the case study reviewed in Section 5—rely on software-level amplitude-embedding routines (e.g., PennyLane’s `AmplitudeEmbedding`) whose classical preprocessing cost is not charged against the “quantum” resource budget, a simplification that must be kept in mind when extrapolating simulator-based results to real hardware.

4.2 Variational Quantum Circuits

Given an encoded input state $|\psi(x)\rangle$, a VQC applies a parametrized unitary $U(\theta)$ —the *ansatz*—and measures an observable M (typically a Pauli operator) to produce a scalar prediction,

$$f(x, \theta) = \langle \psi(x) | U^\dagger(\theta) M U(\theta) | \psi(x) \rangle \in [-1, 1]. \quad (6)$$

When $M = Z_0$ (the Pauli- Z operator on the first qubit), this quantity is linearly remapped to a pseudo-probability,

$$\sigma(f) = \frac{f + 1}{2} \in [0, 1], \quad (7)$$

which can be thresholded for binary classification. A common and hardware-efficient ansatz design alternates layers of single-qubit rotations with a fixed entangling pattern: each layer $\ell = 1, \dots, L$ applies a general single-qubit rotation $\text{Rot}(\phi, \theta, \omega)$ to every qubit, followed by a ring of CNOT gates connecting qubit w to qubit $(w + 1) \bmod n$. For n qubits and L layers, this yields

$$P_{\text{VQC}} = 3nL \quad (8)$$

trainable parameters—e.g., $3 \times 6 \times 3 = 54$ parameters for a 6-qubit, 3-layer circuit, versus the 2,113 parameters of a comparable classical dense head (Equation 4). Gradients of $f(x, \theta)$ with respect to θ can be computed on real hardware via the parameter-shift rule, or, in simulation, via reverse-mode automatic differentiation [20].

4.3 Quantum Convolutional Neural Networks

Quantum convolutional neural networks (QCNNs) translate the locality and parameter sharing of classical CNNs into the quantum circuit formalism. The archetypal QCNN design, introduced by Cong et al. [5] and adapted for NISQ hardware by Wei et al. [36], alternates *quantum convolutional layers*—parametrized two-qubit gates applied to neighboring qubit pairs—with *quantum pooling layers* that halve the active qubit register via measurements and classically controlled single-qubit operations, mimicking the spatial downsampling of classical max-pooling. A key theoretical result is that QCNNs built on translational symmetry are free from the barren-plateau problem under local cost functions, providing a trainability guarantee absent from more generic parametrized circuits [25].

In the context of medical image classification, Khatoniar et al. [15] applied QCNN variants to CXR pneumonia classification under extreme feature reduction, reporting competitive robustness relative to VQC baselines of comparable depth. Wei et al. [36] demonstrated digit recognition and image filtering tasks using a hardware-implementable QCNN whose parameter count is *independent of input size* (determined only by circuit depth and qubit count), an attractive property for deployment on resource-constrained platforms.

It is important to distinguish QCNNs from *quanvolutional* layers [11], in which a small fixed or parametrized quantum circuit processes local image patches in a sliding-window fashion, analogously to a classical convolutional kernel, before passing the resulting feature map to a standard classical CNN. Quanvolutional layers add a quantum pre-processing stage rather than replacing the classical network, and are primarily motivated by the conjecture that quantum circuits can generate feature transformations qualitatively distinct from those achievable by classical filters.

4.4 Quantum Kernel Methods

Quantum kernel methods provide an alternative to the parametrized-circuit paradigm by using a quantum circuit exclusively to evaluate a kernel function,

$$k(x, x') = |\langle \psi(x) | \psi(x') \rangle|^2, \quad (9)$$

and delegating classification to a classical support vector machine (SVM) or kernel ridge regressor trained on the resulting kernel matrix. Havlíček et al. [9] showed, using the ZZ feature map, that the corresponding quantum kernel can be classically hard to estimate exactly, providing a theoretical grounding for potential quantum advantage. Schuld and Killoran [30] established the connection between quantum models and kernel methods more broadly, showing that every VQC implicitly defines a kernel via its state overlap.

For medical imaging, the high dimensionality of image-derived feature vectors poses a practical challenge: evaluating the kernel matrix over a training set of N_{tr} samples requires $O(N_{\text{tr}}^2)$ circuit executions, each querying the state overlap of two encoded feature vectors. Haug et al. [8] proposed scalable quantum kernel estimation via randomized measurements, reducing both the circuit depth required and the sensitivity to noise, and demonstrated large-scale kernel estimation on IBM quantum hardware with noise mitigation. In a comparative hardware study, angle encoding achieved the best overall kernel performance on real QPUs among the encodings tested, albeit with pronounced run-to-run variability, while IQP feature maps performed more consistently yet underperformed relative to classical kernels; dynamical decoupling error mitigation did not uniformly improve performance and was encoding-dependent [27], underscoring that hardware-aware encoding selection is essential rather than optional.

4.5 The Barren-Plateau Problem

The central obstacle to training deep or highly entangling VQCs is the *barren-plateau* phenomenon: for a broad class of randomly initialized, sufficiently expressive ansätze evaluated against a global cost function, the variance of the cost-function gradient with respect to any single parameter θ_i decays exponentially with the number of qubits n ,

$$\text{Var}[\partial_{\theta_i} C(\theta)] = O(2^{-n}), \quad (10)$$

so that, in expectation, gradient-based optimization becomes exponentially difficult to distinguish from noise as n grows [19]. Subsequent work has elucidated the mechanisms underlying this concentration. Cerezo et al. [4] showed that the onset of barren plateaus depends on the *locality* of the cost function: global cost functions—those defined via an operator acting non-trivially on all qubits—are markedly more prone to barren plateaus than local ones. Ortiz Marrero et al. [18] demonstrated that excessive (volume-law) entanglement in the circuit is sufficient to induce barren plateaus independently of the specific ansatz structure. Kashif and Al-Kuwari [13] developed a unified framework showing that barren-plateau onset depends jointly on data encoding, ansatz expressibility, and entanglement, providing design constraints for HQC networks.

These theoretical results have motivated several practical mitigation strategies:

- **Local cost functions and shallow entangling ansätze.** Restricting measurements to local Pauli observables and using nearest-neighbor entangling layers (e.g., the CNOT ring in Equation 8) substantially suppresses gradient variance [4].
- **Controlled initialization.** Initializing parameters near the identity, or near simple product states (“identity blocks”), rather than uniformly at random, ensures that the optimization starts in a region of non-vanishing gradient [19, 23]. Partitioning the circuit into cost and non-cost registers and using low-entanglement initializations provides a practical recipe [23].
- **Expressibility-aware circuit design.** Reducing circuit expressibility—via state-efficient ansätze (SEA) [17] or neural-network-based parameter generation that smooths the loss landscape [38]—trades representational power for trainability.
- **Noise-assisted mitigation.** Entanglement regularization, the addition of controlled Langevin noise, and rotations into the cost function’s eigenbasis can improve convergence on difficult optimization landscapes [23].
- **Noise mitigation as expressibility reduction.** Ding et al. [7] showed that adapting circuit expressibility to match the noise level of the hardware can act as an error-aware mitigation strategy, improving classification accuracy in noisy simulations for QCNNs.

These design constraints guided the layer-count selection for the case-study ansatz in Section 5: three data re-uploading layers on six qubits keep the circuit in a regime of moderate expressibility and shallow entanglement, where gradient-variance concentration is not expected to dominate training on a noise-free simulator.

4.6 On the Prospect of Quantum Advantage

A recurring question in the QML literature is whether VQC-based classifiers can offer a genuine *computational* advantage—in runtime, sample complexity, or expressivity—over classical models, as opposed to merely a reduction in the number of trainable parameters. At the circuit sizes considered in this review (single-digit to low tens of qubits), no such advantage has been established for medical image classification: circuits of this size are efficiently simulable classically, and the reduction in decision-layer parameter count does not, by itself, imply reduced overall computational cost, since the classical feature-extraction and dimensionality-reduction stages typically dominate the total pipeline cost. Claims of quantum advantage in QML more broadly remain an active and contested research area, further complicated by the input/output bottleneck discussed above [1, 2]. The practical value of hybrid VQC decision layers at present should therefore be understood primarily in terms of *parameter efficiency* and *memory footprint*—properties that may be relevant for edge or resource-constrained deployment once fault-tolerant or larger-scale NISQ hardware becomes available—rather than as evidence of an accuracy or speed advantage over well-tuned classical models.

5 Hybrid Quantum–Classical Architectures

5.1 Design Rationale

Given the qubit and depth constraints of NISQ hardware (Section 4) and the high dimensionality of raw medical images, a natural design principle is to separate *feature learning*, delegated to a classical CNN operating on the full-resolution image, from *decision making*, delegated to a compact quantum circuit operating on a low-dimensional, information-dense representation. This principle is shared by all hybrid architectures reviewed below, differing primarily in (i) the choice of classical backbone, (ii) the dimensionality reduction method, (iii) the encoding and

ansatz design, and (iv) whether the quantum component serves as a classifier head, a convolutional feature transformer, or a kernel function.

5.2 Related Work on Hybrid Quantum–Classical CXR Classifiers

Table 1 summarizes published hybrid QML studies targeting chest X-ray analysis. Three themes emerge from this survey.

Table 1: Representative hybrid quantum–classical studies on CXR or pulmonary imaging tasks. All results are from noise-free simulators unless marked with † (IBM Q backend). “HQC-VQC” denotes a classical feature extractor followed by a VQC classifier head; “HQC-Conv” denotes a quanvolutional or QCNN layer integrated into the CNN pipeline.

Reference	Architecture	Dataset	Task	Key result
Kulkarni et al. [14]	HQC-VQC	CXR (Guangzhou)	Pneumonia	Stat. significant improvement vs. classical CNN baseline
Rao et al. [28]	HQC-VQC (†)	COVID-19 Radiography Dataset (15K)	4-class	98.1% test acc.; IBM Q QASM deployment
Decoodt et al. [6]	HQC-VQC	CheXpert (2.4K PA)	Cardiomegaly	$AUC \leq 0.93$; improved Grad-CAM++ localization
Sharma [34]	HQC-Conv	Custom CXR	Pneumonia	ZFNet + quantum layer hybrid
Khatoniar et al. [15]	QCNN	CXR (Guangzhou)	Pneumonia	Robustness analysis under extreme feature reduction
This work	HQC-VQC	CXR (Guangzhou, 5.8K)	Pneumonia	AUC 0.860 (VQC) vs. 0.940 (MLP); $39\times$ param. reduction

Consistent accuracy–efficiency trade-off. All surveyed studies confirm the qualitative finding that a hybrid quantum component can achieve meaningful classification performance at a substantially reduced decision-layer parameter budget. However, results that appear favorable relative to their specific classical baselines often reflect comparisons against purposely weak baselines rather than state-of-the-art classical models [1].

Variability in experimental rigor. Dataset sizes range from a few hundred to $\sim 15,000$ images; train/test split rationale is frequently underspecified; and class prevalence in the test set is rarely reported explicitly. This limits direct cross-study comparison and motivates the shift-aware evaluation framework presented in Section 7.

Limited hardware deployment. Only Rao et al. [28] report partial deployment on a real IBM Q backend (QASM simulator), and no study provides a systematic comparison of simulator versus hardware performance under noise. The gap between simulator and hardware results—which Section 6 is designed to quantify—remains the most critical open empirical question in the field.

5.3 Pipeline Architecture

The pipeline consists of four stages, summarized in Table 2: (1) a frozen ConvNeXt-Tiny backbone, used headless, maps each 224×224 input image to a 768-dimensional embedding; (2) a

domain-adversarial neural network (DANN) with a gradient reversal layer (GRL) is trained on top of the embedding to suppress the dataset shift between the training and test distributions (Section 5.2); (3) a nonlinear autoencoder, fitted on the training embeddings, reduces this embedding to 64 dimensions, chosen to match the Hilbert-space dimension of a 6-qubit register ($2^6 = 64$); and (4) a 6-qubit, 3-layer VQC, using amplitude encoding (Equation 5) and the rotation-plus-ring-CNOT ansatz of Section 4.2, produces a scalar pneumonia score via Equations 6–7.

Table 2: Hybrid pipeline architecture and parameter counts.

Stage	Input	Operation	Output	Parameters
Feature extraction	224×224 image	ConvNeXt-Tiny (frozen)	768-D vector	0 (fixed)
Domain adaptation	768-D	DANN + GRL	768-D vector	0 (fixed)
Dimensionality reduction	768-D	Autoencoder (nonlinear)	64-D vector	0 (fixed)
Quantum classification	64-D	VQC $U(\theta)$	score $\in [0, 1]$	54

Among linear PCA, linear discriminant analysis (LDA), and univariate feature selection (SelectKBest), a nonlinear autoencoder yielded the most stable training dynamics and the best validation performance at 64 latent dimensions; LDA suffered from numerical instability in the high-dimension/limited-sample regime, and SelectKBest achieved marginally weaker validation metrics. The autoencoder therefore replaces the linear projection used in earlier variants of this pipeline.

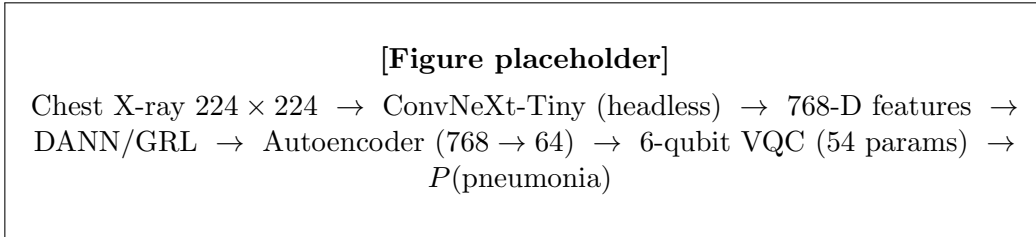


Figure 1: Schematic of the hybrid CNN–autoencoder–VQC pipeline.

5.4 Dataset and Preprocessing

The pipeline was evaluated on the publicly available pediatric CXR pneumonia dataset originating from the Guangzhou Women and Children’s Medical Center (5,856 anteroposterior CXR images, ages approximately 1–5 years). The original release provides a validation split of only 16 images, which is insufficient for reliable convergence monitoring; the training and validation partitions were therefore merged and re-split 80:20 with class stratification, while the original test partition was left untouched to preserve the integrity of the final evaluation. The resulting split is summarized in Table 3. Notably, the test set exhibits a materially different class prevalence (62.5% pneumonia) than the training and validation sets (74.2% pneumonia), constituting an explicit, quantified dataset shift that is exploited below as a robustness stress test rather than concealed.

Standard image preprocessing was applied prior to the ConvNeXt-Tiny forward pass: resizing to 224×224 pixels, conversion to a normalized float tensor, and channel-wise standardization using ImageNet mean $[0.485, 0.456, 0.406]$ and standard deviation $[0.229, 0.224, 0.225]$, matching the statistics of the pretraining distribution. Because the CXR images are grayscale while the backbone expects three input channels, the single channel is duplicated across the RGB channels. Medical-safe augmentations were limited to mild rotations ($\pm 7^\circ$), small affine translations,

Table 3: Dataset composition and class-prevalence shift.

Split	Images	Pneumonia (%)	Normal (%)
Training	4,185	74.2	25.8
Validation	1,047	74.2	25.8
Test	624	62.5	37.5

and modest color jitter; aggressive augmentations such as horizontal flipping and RandAugment were deliberately excluded as they are harmful for anatomical X-ray content.

5.5 Training Configuration and Loss Functions

The VQC and a classical baseline classifier (a single-hidden-layer MLP with 32 hidden units, operating on the same 64-dimensional autoencoder features) were trained independently, each with the Adam optimizer, a batch size of 16, and early stopping (patience of 3 epochs) on validation loss. Because VQC outputs (Equation 6) lie in $[-1, 1]$ rather than $[0, 1]$, binary labels $y_i \in \{0, 1\}$ were remapped to targets $t_i \in \{-1, +1\}$, and the quantum classifier was trained with a class-weighted mean squared error,

$$\mathcal{L}_{\text{QMSE}} = \frac{1}{N} \sum_{i=1}^N w(t_i) (z_i - t_i)^2, \quad (11)$$

where z_i is the raw circuit output and $w(\cdot)$ up-weights the minority (Normal) class to counteract the training-set prevalence imbalance. The classical baseline was trained with a class-weighted binary cross-entropy loss on its sigmoid output. The quantum classifier’s learning rate followed a linear warm-up over the first $E_{\text{warmup}} = 3$ epochs followed by cosine annealing,

$$\eta(e) = \begin{cases} \eta_0 \frac{e}{E_{\text{warmup}}}, & e \leq E_{\text{warmup}}, \\ \eta_{\min} + \frac{1}{2}(\eta_0 - \eta_{\min}) \left[1 + \cos\left(\pi \frac{e - E_{\text{warmup}}}{E - E_{\text{warmup}}}\right) \right], & e > E_{\text{warmup}}, \end{cases} \quad (12)$$

with $\eta_0 = 10^{-3}$ and $\eta_{\min} = 10^{-5}$, while the classical baseline used unmodified cosine annealing. Gradient computation for the quantum circuit used the adjoint differentiation method on the simulator, which is dramatically faster than the parameter-shift rule and requires no additional circuit executions beyond the forward pass. All quantum-circuit simulation was performed on a GPU-accelerated statevector simulator (noise-free), so the reported results characterize the idealized, decoherence-free behavior of the ansatz rather than performance on physical NISQ hardware. In practice both models converged well before the 50-epoch budget (early stopping triggered at approximately 7–10 epochs).

5.6 Results

Table 4 summarizes test-set performance for the classical baseline and the hybrid VQC classifier at the validation-selected decision thresholds (Section 7.2): $\tau = 0.30$ for the MLP and $\tau = 0.35$ for the VQC. Both thresholds were selected to maximize balanced accuracy on the validation set only, and applied once to the test set.

The classical baseline outperforms the hybrid quantum classifier on every reported metric, consistent with the expectation that a fully connected head can better exploit a rich, 64-dimensional continuous representation than a shallow, locally entangling 6-qubit circuit. The gap is nevertheless modest—the accuracy difference is 1.28 percentage points and is not statistically significant at $\alpha = 0.05$ under McNemar’s test applied to the 624 paired test predictions—so the hybrid model attains essentially comparable discriminative performance (test AUC-ROC

Table 4: Test-set diagnostic performance: classical MLP baseline vs. hybrid VQC classifier (624 test images, 62.5% pneumonia prevalence). Thresholds τ were selected on the validation set via balanced-accuracy maximization and applied once to the test set.

Metric	Classical (MLP)	Hybrid (VQC)
Accuracy	82.53%	81.25%
Precision	78.27%	77.03%
Sensitivity (Recall)	99.74%	99.74%
Specificity	53.85%	50.43%
F_1 -score	0.8771	0.8693
AUC-ROC (test)	0.940	0.860
Balanced accuracy	0.7679	0.7509
Decision threshold τ	0.30	0.35
Decision-layer parameters	2,113	54
Parameter reduction	$\approx 39\times$	

0.860) using approximately $39\times$ fewer decision-layer parameters (54 versus 2,113). Both models exhibit a “cautious screening” operating profile, with very high sensitivity (99.74%) at the cost of low specificity (50.4–53.9%), which from a purely metric standpoint favors minimizing missed pneumonia cases over minimizing false alarms; we emphasize that no clinical validation of this trade-off is implied. The modest specificity is attributable to the decision thresholds selected by the balanced-accuracy criterion, which deliberately trades specificity for sensitivity in an imbalanced test population.

Compared to the literature surveyed in Table 1, the test AUC-ROC of 0.860 is lower than the near-perfect figures reported by Rao et al. [28] or Kulkarni et al. [14], differences that can be attributed to: (i) the explicit train/test class-prevalence shift applied here; (ii) the single-run evaluation without cross-validation; and (iii) the absence of fine-tuning of the ConvNeXt-Tiny backbone. These factors collectively reduce the upper bound on achievable performance and make the comparison more conservative and, arguably, more realistic.

6 Experiments on Real Quantum Hardware

[PLACEHOLDER — to be completed with real-hardware results]

This section will report the results of executing the VQC classifier (Section 5) on one or more physical NISQ devices (e.g., IBM Quantum superconducting processors or trapped-ion platforms). It will include the following subsections once the experiments are complete.

6.1 Motivation and Hardware Selection

[PLACEHOLDER — to be completed with real-hardware results]

Justification for the choice of hardware platform(s); qubit counts, native gate sets, and typical two-qubit gate error rates of the selected devices; circuit transpilation and optimization strategy (e.g., routing, SWAP insertion, basis-gate decomposition); and the mapping between the 6-qubit VQC ansatz and the device’s connectivity graph.

6.2 Error Mitigation Strategies

[PLACEHOLDER — to be completed with real-hardware results]

Description of the error mitigation protocol(s) to be applied: readout-error calibration, zero-noise extrapolation (ZNE), probabilistic error cancellation (PEC), or dynamical decoupling (DD). Discussion of encoding-dependent mitigation effectiveness in light of the comparative literature [27, 8], and the expected trade-off between mitigation overhead (additional circuit executions) and performance gain.

6.3 Hardware Results and Simulator–Hardware Gap

[PLACEHOLDER — to be completed with real-hardware results]

Test-set performance metrics (Table 4 format) obtained on the physical device, with and without error mitigation. Quantitative comparison against the noise-free statevector simulator results to assess the magnitude of the hardware degradation. Analysis of which circuit components (state preparation, entangling layers, measurement) contribute most to the performance gap, and recommendations for circuit re-design to improve hardware fidelity.

7 Experimental Considerations and Evaluation Metrics

7.1 Metrics

Given the class imbalance typical of public pneumonia CXR datasets, accuracy alone is an insufficient evaluation criterion. We recommend reporting, at minimum: sensitivity (recall), $\text{Sens} = TP/(TP + FN)$; specificity, $\text{Spec} = TN/(TN + FP)$; balanced accuracy, $\text{BalAcc} = \frac{1}{2}(\text{Sens} + \text{Spec})$; the F_1 -score; and the area under the receiver operating characteristic curve (AUC-ROC), which is threshold-independent and therefore particularly informative when comparing classifiers (such as a VQC and an MLP) whose raw outputs have different natural ranges and calibration properties.

7.2 Threshold Selection

Because the VQC output (Equation 7) and the classical sigmoid output are continuous scores, a decision threshold τ must be chosen to obtain a binary prediction $\hat{y} = \mathbb{I}[p_{\text{pneumonia}} > \tau]$. A principled approach is to perform a discrete threshold scan on the validation set, selecting $\tau^* = \arg \max_{\tau} \text{BalAcc}(\tau)$, and to report test-set metrics at this fixed, validation-selected threshold rather than at an arbitrarily chosen value. In the case study reviewed here, this procedure selected $\tau^* = 0.30$ for the MLP and $\tau^* = 0.35$ for the VQC over a scan range of $\tau \in [0.30, 0.80]$ with step 0.025, both below the conventional default of 0.5. The sensitivity analysis itself remains a useful diagnostic: thresholds near the conventional 0.5 default shift the operating point toward higher specificity at the cost of sensitivity, whereas the balanced-accuracy criterion favors the low-threshold, high-sensitivity operating point reported in Table 4.

7.3 Dataset Shift as an Explicit Evaluation Axis

We highlight dataset shift as a first-class evaluation concern rather than an incidental property of the data split. Reporting class prevalence separately for training, validation, and test partitions (as in Table 3) allows a reader to distinguish genuine generalization performance from performance inflated by matching train/test class ratios. This is particularly relevant for QML studies, where sample-efficiency constraints (imposed by simulator run-time or hardware queue

availability) often preclude the large-scale cross-validation protocols that are standard in classical deep learning benchmarking; a single, explicitly characterized shift-aware split is a partial, low-cost mitigation. The generalizability limitations of single-institution, narrow-demographic datasets highlighted by Zech et al. [37] apply with equal force to hybrid quantum–classical architectures.

7.4 Interpretability

For classical CNN components, gradient-based saliency methods such as Grad-CAM can verify that the network attends to clinically relevant lung regions rather than to background artifacts (e.g., imaging equipment, laterality markers), providing a qualitative sanity check complementary to aggregate performance metrics. Decoodt et al. [6] found that augmenting a DenseNet-121 backbone with a variational quantum layer improved Grad-CAM++ localization quality for cardiomegaly detection, suggesting a potential ancillary benefit of quantum components beyond raw classification accuracy. Analogous interpretability tools for the quantum component are considerably less mature; feasible directions include measuring the sensitivity of the circuit output to perturbations of individual input amplitudes, or analyzing feature importance in the classical dimensionality-reduction stage (e.g., via SelectKBest or PCA loading vectors) as a proxy for what information reaches the quantum classifier.

8 Challenges, Limitations, and Open Problems

We summarize the principal limitations of current hybrid VQC approaches to CXR pneumonia detection, several of which are directly evidenced by the case study in Section 5.

1. **Barren plateaus.** As circuit width or entangling depth increases, gradient variance decays exponentially (Equation 10), limiting the ansätze that can be reliably trained with current optimizers [19, 4, 18]. The case-study ansatz was therefore deliberately restricted to three shallow layers on six qubits, a regime in which gradient-variance concentration is not expected to dominate training (Section 4.5). While several mitigation strategies exist, none fully eliminates the fundamental exponential scaling [13].
2. **State-preparation and encoding overhead.** Amplitude encoding achieves exponential qubit compression but, in the general case, incurs exponential state-preparation circuit depth [1]; simulator-based studies typically do not account for this cost, which will materially affect feasibility on physical hardware.
3. **Dataset shift.** Differences in class prevalence (and, more broadly, in acquisition device, patient demographics, and imaging protocol) between training and deployment distributions are a well-documented source of generalization failure for CXR classifiers [37], and were observed to degrade specificity in the reviewed case study.
4. **Hardware noise.** All simulation results reviewed here were obtained on noise-free statevector simulators; real NISQ devices introduce gate errors, readout errors, and decoherence that are expected to degrade performance unless mitigated. Empirical evidence from quantum kernel studies suggests that error mitigation effectiveness is strongly encoding-dependent and that no universally applicable mitigation protocol has emerged [27, 8].
5. **Information loss under dimensionality reduction.** Compressing a 768-dimensional CNN embedding to 64 (or fewer) dimensions via the nonlinear autoencoder necessarily discards variance, some fraction of which may be diagnostically relevant. Alternative encodings that avoid aggressive compression (e.g., data re-uploading [24] with angle encoding, or distributed amplitude encoding over multiple shorter circuits) remain open design questions [33].

6. **Absence of demonstrated quantum advantage.** At the circuit scales considered (single-digit qubits), no computational or statistical advantage over classical models has been established; the case for VQC decision layers currently rests on parameter efficiency rather than accuracy or speed.
7. **Limited statistical robustness.** The case study reviewed here uses a single train/validation/test split without cross-validation or repeated randomized resampling, owing to the computational cost of circuit simulation; reported performance differences, while accompanied by approximate confidence intervals, should be interpreted with the caveats appropriate to a single-run evaluation.
8. **Single-institution, narrow-demographic data.** The underlying dataset originates from a single pediatric population at one medical center, limiting claims of generalizability to adult populations, other acquisition protocols, or other institutions.
9. **Unfair classical baselines.** Several studies surveyed in Table 1 compare a quantum model favorably against a deliberately weak classical reference, inflating the apparent advantage of the hybrid approach. Rigorous evaluation requires comparison against optimized classical baselines of equal or greater parameter budget on the quantum side [1].

9 Future Directions

Several directions appear promising for advancing hybrid quantum–classical pneumonia detection beyond the proof-of-concept stage.

- **Execution on physical quantum hardware.** Migrating the VQC component to a cloud-accessible NISQ device (e.g., superconducting or trapped-ion platforms), combined with hardware-aware error mitigation [27, 8], will quantify the simulator-to-hardware performance gap and is a necessary step before any hardware-relevant advantage claim can be assessed. This is the primary goal of Section 6 (forthcoming).
- **Quantum convolutional and quanvolutional architectures.** QCNNs [5, 36] that exploit spatial locality in the quantum circuit, and quanvolutional pre-processing layers [11] that apply quantum filters to image patches prior to classical CNN processing, represent theoretically motivated alternatives to the “quantum classifier head” paradigm. Their trainability guarantees [25] and parameter independence from input size make them particularly attractive for deployment under NISQ constraints.
- **Quantum kernel methods.** Replacing the VQC classifier with a quantum kernel SVM [9, 30] has the advantage of convex optimization—eliminating barren plateaus by construction—and a well-defined notion of kernel quality via the quantum Fisher information [8]. Scalable kernel estimation via randomized measurements [8] makes this approach computationally feasible even for training sets of several thousand images.
- **Data re-uploading classifiers.** Applying the data re-uploading paradigm [24] within the VQC decision layer would increase circuit expressibility without requiring additional qubits, potentially alleviating the information-loss constraints imposed by aggressive dimensionality reduction to the 64-dimensional amplitude-encoded register.
- **Systematic hyperparameter optimization.** Bayesian optimization frameworks (e.g., Optuna) applied jointly across classical and quantum hyperparameters (learning rate, ansatz depth, entangling topology, number of retained latent dimensions) would allow more efficient exploration of the design space than manual or grid search, particularly given the high per-run cost of circuit simulation.

- **Rigorous, shift-aware, multi-institutional evaluation.** Extending evaluation to larger, multi-site datasets (e.g., CheXpert, NIH ChestX-ray14) and explicit domain-adaptation techniques (reweighting, adversarial domain adaptation) would provide a more realistic assessment of generalization than single-institution pediatric data [37].
- **Ensemble and hybrid-decision strategies.** Combining classical and quantum model outputs (e.g., via weighted logit averaging or a meta-classifier) could exploit differences in the error profiles of the two approaches, potentially improving both sensitivity and specificity relative to either model alone [16].
- **Quantum-side interpretability.** Developing sensitivity- and perturbation-based tools analogous to Grad-CAM for the quantum decision layer would help clarify whether the VQC learns complementary or redundant information relative to the classical autoencoder projection, and would support the broader goal of building trustworthy clinical decision-support systems.

10 Conclusion

This paper reviewed the computational and algorithmic foundations of hybrid quantum–classical architectures for pneumonia detection from chest X-ray images, with an emphasis on data-encoding strategies, variational and convolutional ansatz design, quantum kernel methods, and the trainability limitations imposed by barren plateaus together with their principal mitigation strategies. We surveyed a growing body of literature applying hybrid QML pipelines to CXR tasks—including recent works demonstrating statistically significant improvements over classical baselines [14] and near-unity accuracy on COVID-19 radiography datasets [28]—while noting the evaluation limitations that temper direct comparison across studies.

Using a ConvNeXt-Tiny/autoencoder/VQC pipeline evaluated on a public pediatric CXR dataset as a concrete case study, we showed that a 6-qubit, 54-parameter variational quantum classifier can achieve a level of diagnostic performance comparable to—though not exceeding—that of a classical MLP of comparable input dimensionality (test AUC-ROC 0.860 versus 0.940; accuracy 81.25% versus 82.53%), while requiring approximately $39\times$ fewer trainable parameters in the decision layer (54 versus 2,113). This result should be read narrowly: it demonstrates that a compact variational circuit can serve as a functional, parameter-efficient classifier on a compressed medical-imaging representation, not that a computational or statistical quantum advantage has been achieved. Barren plateaus, state-preparation overhead, aggressive dimensionality-reduction-induced information loss, and dataset shift all constrain current performance, and the simulation-based results reported here have yet to be validated on physical quantum hardware.

We regard hybrid VQC decision layers, quantum convolutional architectures, and quantum kernel classifiers as promising but early-stage research directions whose eventual clinical or practical relevance will depend on: progress in fault-tolerant or higher-fidelity NISQ hardware; more expressive yet trainable ansatz designs (e.g., data re-uploading, low-entanglement initialization); hardware-aware error mitigation; and substantially more rigorous, multi-institutional, shift-aware evaluation protocols than are currently standard in the literature. The forthcoming hardware experiments (Section 6) will provide the first direct empirical quantification of the simulator-to-hardware gap for this specific pipeline and will constitute a necessary precondition for any hardware-grounded advantage claim.

References

- [1] Aaronson, S. (2015). Read the fine print. *Nature Physics*, 11(4), 291–293.

- [2] Biamonte, J., Wittek, P., Pancotti, N., Rebentrost, P., Wiebe, N., & Lloyd, S. (2017). Quantum machine learning. *Nature*, 549(7671), 195–202.
- [3] Centers for Disease Control and Prevention. (2025). *About Pneumonia*. <https://www.cdc.gov/pneumonia/about/index.html>
- [4] Cerezo, M., Sone, A., Volkoff, T., Cincio, L., & Coles, P. J. (2021). Cost-function-dependent barren plateaus in shallow parametrized quantum circuits. *Nature Communications*, 12(1), 1791.
- [5] Cong, I., Choi, S., & Lukin, M. D. (2019). Quantum convolutional neural networks. *Nature Physics*, 15(12), 1273–1278.
- [6] Decoodt, P., Liang, T. J., et al. (2023). Hybrid classical–quantum transfer learning for cardiomegaly detection in chest X-rays. *Journal of Imaging*, 9(7), 128. <https://doi.org/10.3390/jimaging9070128>
- [7] Ding, Y., Zhang, S., & Li, X. (2024). Noise mitigation: an efficient approach for quantum convolutional neural networks to enhance performance in NISQ. Preprint. <https://doi.org/10.21203/rs.3.rs-4594670/v1>
- [8] Haug, T., Self, C. N., & Kim, M. (2021). Quantum machine learning of large datasets using randomized measurements. *arXiv preprint arXiv:2108.01039*.
- [9] Havlíček, V., Córcoles, A. D., Temme, K., Harrow, A. W., Kandala, A., Chow, J. M., & Gambetta, J. M. (2019). Supervised learning with quantum-enhanced feature spaces. *Nature*, 567(7747), 209–212.
- [10] He, K., Zhang, X., Ren, S., & Sun, J. (2016). Deep residual learning for image recognition. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 770–778.
- [11] Henderson, M., Shakya, S., Pradhan, S., & Cook, T. (2020). Quanvolutional neural networks: Powering image recognition with quantum circuits. *Quantum Machine Intelligence*, 2(1), 2.
- [12] Institute for Quality and Efficiency in Health Care (IQWiG). (2021). *Overview: Pneumonia*. InformedHealth.org. <https://www.ncbi.nlm.nih.gov/books/NBK525774/>
- [13] Kashif, M., & Al-Kuwari, S. (2023). The unified effect of data encoding, ansatz expressibility and entanglement on the trainability of HQNNs. *International Journal of Parallel, Emergent and Distributed Systems*, 38(5), 362–400. <https://doi.org/10.1080/17445760.2023.2231163>
- [14] Kulkarni, V., Pawale, S., & Kharat, A. (2023). A classical–quantum convolutional neural network for detecting pneumonia from chest radiographs. *Neural Computing and Applications*, 35(21), 15503–15510. <https://doi.org/10.1007/s00521-023-08566-1>
- [15] Khatoniar, R., Vyas, V., Acharya, V., et al. (2024). Quantum convolutional neural network for medical image classification. In *Proceedings of the 2nd International Conference on Trends in Quantum Computing and Emerging Business Technologies (TQCEBT)*. IEEE. <https://doi.org/10.1109/tqcebt59414.2024.10545121>
- [16] Kundu, R., Das, R., Geem, Z. W., Han, G.-T., & Sarkar, R. (2021). Pneumonia detection in chest X-ray images using an ensemble of deep learning models. *PLOS ONE*, 16(9), e0256630.

- [17] Liu, X., Liu, G., Huang, J., et al. (2022). Mitigating barren plateaus of variational quantum eigensolvers. *arXiv preprint arXiv:2205.13539*.
- [18] Ortiz Marrero, C., Kieferová, M., & Wiebe, N. (2021). Entanglement-induced barren plateaus. *PRX Quantum*, 2(4), 040316. <https://doi.org/10.1103/PRXQuantum.2.040316>
- [19] McClean, J. R., Boixo, S., Smelyanskiy, V. N., Babbush, R., & Neven, H. (2018). Barren plateaus in quantum neural network training landscapes. *Nature Communications*, 9(1), 4812.
- [20] Mitarai, K., Negoro, M., Kitagawa, M., & Fujii, K. (2018). Quantum circuit learning. *Physical Review A*, 98(3), 032309.
- [21] Naz, A. (2020). Literature review on X-ray based pneumonia detection using deep learning. *LCJSTEM*, 1(1), 1–9.
- [22] Nielsen, M. A., & Chuang, I. L. (2010). *Quantum Computation and Quantum Information* (10th anniversary ed.). Cambridge University Press.
- [23] Patti, T. L., Najafi, K., Gao, X., & Yelin, S. F. (2021). Entanglement devised barren plateau mitigation. *Physical Review Research*, 3, 033090. <https://doi.org/10.1103/PhysRevResearch.3.033090>
- [24] Pérez-Salinas, A., Cervera-Lierta, A., Gil-Fuster, E., & Latorre, J. I. (2020). Data re-uploading for a universal quantum classifier. *Quantum*, 4, 226.
- [25] Pesah, A., Cerezo, M., Wang, S., Volkoff, T., Sornborger, A. T., & Coles, P. J. (2021). Absence of barren plateaus in quantum convolutional neural networks. *Physical Review X*, 11(4), 041011.
- [26] Preskill, J. (2018). Quantum computing in the NISQ era and beyond. *Quantum*, 2, 79.
- [27] Beaulieu, D., Miracle, D., Pham, A. T., & Scherr, W. (2022). Quantum kernel for image classification of real-world manufacturing defects. *arXiv preprint arXiv:2212.08693*.
- [28] Rao, G. V. E., Rajitha, B., Srinivasu, P. N., Ijaz, M. F., & Woźniak, M. (2024). Hybrid framework for respiratory lung diseases detection based on classical CNN and quantum classifiers from chest X-rays. *Biomedical Signal Processing and Control*, 88, 105567. <https://doi.org/10.1016/j.bspc.2023.105567>
- [29] Schuld, M., & Petruccione, F. (2018). *Supervised Learning with Quantum Computers*. Springer, Quantum Science and Technology.
- [30] Schuld, M., & Killoran, N. (2019). Quantum machine learning in feature Hilbert spaces. *Physical Review Letters*, 122(4), 040504.
- [31] Schuld, M. (2021). Supervised quantum machine learning models are kernel methods. *arXiv preprint arXiv:2101.11020*.
- [32] Ortiz-Toro, C., García-Pedrero, A., Lillo-Saavedra, M., & Gonzalo-Martín, C. (2022). Automatic detection of pneumonia in chest X-ray images using textural features. *Computers in Biology and Medicine*, 145, 105466. <https://doi.org/10.1016/j.compbiomed.2022.105466>
- [33] Sharma, A., et al. (2023). Survey of encoding techniques for quantum machine learning. *Cybernetics and Physics*, 12(1), 45–67.

- [34] Sharma, R. (2023). Classification of pneumonia via a hybrid ZFNet–quantum neural network. *Current Medical Imaging Reviews*. Advance online publication.
- [35] Sim, S., Johnson, P. D., & Aspuru-Guzik, A. (2019). Expressibility and entangling capability of parametrized quantum circuits for hybrid quantum-classical algorithms. *Advanced Quantum Technologies*, 2(12), 1900070.
- [36] Wei, S., Chen, Y., Zhou, Z.-R., & Long, G.-L. (2022). A quantum convolutional neural network on NISQ devices. *AAPPS Bulletin*, 32(1), 2. <https://doi.org/10.1007/s43673-021-00030-3>
- [37] Zech, J. R., Badgeley, M. A., Liu, M., Costa, A. B., Titano, J. J., & Oermann, E. K. (2018). Variable generalization performance of a deep learning model to detect pneumonia in chest radiographs: A cross-sectional study. *PLOS Medicine*, 15(11), e1002683.
- [38] Zhu, Y., Liang, Y., & Situ, H. (2024). Enhancing variational quantum circuit training: an improved neural network approach for barren plateau mitigation. *arXiv preprint arXiv:2411.09226*.